

Juan Pablo Villegas Ceballos

Electronic Engineer, M.Sc. in Automation and Industrial Control, Ph.D. Candidate in Engineering – Automation, and Associate Researcher recognized by Minciencias. Specialist in power electronics, control of switched-mode converters, renewable energy integration, and educational hardware.

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Education

Universidad Nacional de Colombia , Engineering Focus on advanced control, energy management systems, and microgrids.	Medellín, Colombia Jan 2022 – Dec 2026
Instituto Tecnológico Metropolitano (ITM) , Automation and Industrial Control Thesis: Optimization methodology for sliding-mode control of energy storage systems to extend battery lifespan.	Medellín, Colombia Jan 2016 – Jan 2019
Instituto Tecnológico Metropolitano (ITM) , Electronic Engineering - Professional License AN206-105671. - Thesis: Design and implementation of a vector control system for brushless DC motors applied to robotics.	Medellín, Colombia Jan 2012 – Jan 2014
Instituto Tecnológico Metropolitano (ITM) , Electronic Technology Thesis: Scanning sonar for autonomous robotic agent navigation.	Medellín, Colombia Jan 2008 – Jan 2011

Experience

Instituto Tecnológico Metropolitano (ITM) , Assistant Professor / Docente Ocasional Teaching undergraduate courses in Power Electronics, Industrial Electronics, Continuous and Discrete Control Systems, and Control of Switched Converters. - Design of Hardware-in-the-Loop (HIL) systems for educational and research labs. - Research lead in renewable energy, microgrids, and power converter design.	Medellín, Colombia Jan 2020 – Dec 2026 7 years
Institución Universitaria Pascual Bravo , Adjunct Professor / Researcher Taught courses in Applied Software, Servomechanisms, and Electronic Labs. Led R&D activities in renewable energy.	Medellín, Colombia Jan 2018 – Jan 2019 1 year 1 month
Trazabilidad Global S.A.S. , Hardware & Software Integration Engineer Developed embedded GTPC solutions for public transit telemetry and automated weighing platforms for agro-industrial applications.	Medellín, Colombia Jan 2014 – Jan 2017 3 years 1 month

Awards

Honourable Mention Institutional award granted for outstanding research and academic contribution. Instituto Tecnológico Metropolitano	Apr 2026
Honorable Mention for Research Work Granted by Resolution No. 529 for outstanding products developed within research seedbeds. Consejo de la Facultad de Ingenierías, ITM	Sept 2012

Publications

Battery Supporting device to reduce current transients: Co-design of the power stage and an adaptive control

Co-design of a Cuk-based power converter and adaptive sliding mode controller to mitigate battery degradation in hybrid storage systems.

Juan Pablo Villegas Ceballos, Carlos Andrés Ramos Paja, Daniel González Montoya, Elkin Edilberto Henao Bravo, Andrés Julián Saavedra Montes, Sergio Ignacio Serna Garcés

doi.org/10.1016/j.rineng.2025.105083

Cost-effective and versatile Hardware-in-the-Loop system for DC/DC converter emulation in education and research

Development of an open-hardware HIL setup designed for emulating switching converters in academic power electronics courses.

Juan Pablo Villegas Ceballos, Alejandro Gaviria Cano, Cristian Escudero Quintero, José David López Suárez, Elkin Edilberto Henao Bravo, Sergio Ignacio Serna Garcés

doi.org/10.1016/j.ohx.2025.e00701

El transporte férreo y fluvial colombiano: prospectiva hacia la electromovilidad

Research book evaluating the penetration and impact of electromobility in Colombian river and railway transportation networks.

Juan Pablo Villegas Ceballos, Daniel González Montoya, et al.

Skills

Power Electronics & Control

Programming & Tools

Languages

Spanish

Native speaker

English

B2 - Professional Working Proficiency

Interests

Research & Pedagogy